RESEARCH DOCUMENTATION

Explainable Artificial Intelligence (XAI) in Education

Uses, Applications, and Recommendations for Teachers

Systematic Review of the Scientific Literature, 2019–2025

Based on 35+ primary studies | IEEE, Springer, MDPI, ACM, arXiv

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Abstract

Explainable Artificial Intelligence (XAI) refers to methods and tools that render the decision-making processes of AI systems transparent and understandable to human users. In the educational context, XAI emerges as both a pedagogical and ethical necessity: AI models can predict student failure, but without explanation, neither the teacher nor the student can understand which factors led to that prediction, nor act upon it. This documentation synthesises recent research (2019–2025) on five major application domains of XAI in education: performance prediction, dropout prevention, intelligent tutoring systems, automated essay scoring, and adaptive learning. Drawing on over 35 empirical studies and systematic reviews, it examines the main XAI techniques (SHAP, LIME, Anchors, Counterfactuals), their documented benefits, their ethical challenges, and their practical limitations. The document concludes with ten concrete, evidence-based recommendations for teachers and institutions deploying AI-powered educational tools.

Keywords: explainable AI, XAI, education, learning analytics, SHAP, LIME, student prediction, intelligent tutoring systems, algorithmic bias, teacher trust.

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1. Introduction

1.1 Context and Problem Statement

The integration of artificial intelligence into educational systems has grown substantially over the past decade. From adaptive learning platforms to automated grading tools and recommendation systems, AI offers unprecedented possibilities for personalisation at scale. Yet this expansion raises a fundamental question identified by *Turkmen (2025)*[1] in a systematic review of 35 studies: most AI algorithms operate as *black boxes* — they produce results without exposing their reasoning. A system may predict that a student will fail an examination, but neither the teacher nor the student can understand which factors drove that prediction.

This opacity creates a dual problem. First, it erodes the **trust** of educational stakeholders in AI tools. Second, it prevents **targeted pedagogical intervention**: if the teacher does not know which levers to act on, a risk alert is of limited value. The absence of explanation also raises serious ethical concerns around accountability and the potential for algorithmic discrimination.

1.2 Definition of XAI

Explainable Artificial Intelligence (XAI) refers to systems that make AI models more transparent, helping users understand how outputs are generated.[1] *Altukhi and Pradhan (2025)*[7], in a PRISMA-based systematic review, identified 15 distinct definitions and 62 challenges associated with XAI in educational contexts, revealing the absence of terminological consensus. For the purposes of this document, we adopt the widely cited definition of *Doshi-Velez and Kim*: "the ability to explain or present AI/ML systems in understandable terms to a human" — a definition with over 5,000 citations as of 2024.[9]

It is important to distinguish **interpretability** (a property of the model itself) from **explainability** (the ability to produce post-hoc explanations of an opaque model). Both concepts are relevant to education, as different stakeholders — teachers, students, administrators, parents — require different levels and formats of explanation.

1.3 Scope and Methodology

This document is based on a review of scientific literature published between 2019 and 2025, indexed in IEEE Xplore, Springer Nature, MDPI, ACM Digital Library, and arXiv. Search terms included: "Explainable AI", "XAI", "education", "learning analytics", "student prediction", "SHAP", "LIME", "intelligent tutoring system", "dropout prediction". The selection prioritises systematic reviews, empirical studies with human participants, and theoretical frameworks with direct implications for pedagogical practice. Studies focusing purely on technical algorithm comparisons without educational implications were excluded.

2. XAI Techniques: An Overview

2.1 Intrinsic vs. Post-Hoc Explainability

The literature distinguishes two broad families of XAI methods according to when explainability is produced.[5]

Intrinsic explainability (ante-hoc) refers to models that are inherently interpretable: decision trees, logistic regression, rule-based systems. These models offer native transparency but are often less performant when dealing with the complex, high-dimensional data typical of educational settings.

Post-hoc explainability applies to already-trained complex models (neural networks, random forests, gradient boosting) and seeks to explain their decisions retrospectively. This is the category that contains the most widely used techniques in educational AI research. A key debate in the field concerns the faithfulness of such explanations: does a post-hoc explanation truly reflect the model's internal reasoning, or does it produce a plausible but inaccurate approximation?

2.2 SHAP — SHapley Additive exPlanations

SHAP, grounded in cooperative game theory (Shapley values), has become the dominant XAI method in educational data mining.[11] Its core principle: each input variable (engagement rate, attendance, assignment scores, submission timing) is treated as a "player" and its contribution to the final prediction is computed by considering all possible variable combinations. This yields a mathematically grounded attribution value for each feature.

SHAP provides both **global explanations** (which variables matter most across all students?) and **local explanations** (why does the model predict that this specific student will fail?). This dual granularity is particularly useful for pedagogical intervention. An empirical study using the Open University Learning Analytics Dataset (OULAD) found that SHAP-augmented models achieved up to **94% accuracy** in identifying at-risk students, while revealing that early engagement and registration timing are the most determinant predictors.[18]

A limitation of SHAP, noted by *Salih et al. (2025)*[15], is its sensitivity to feature collinearity: when input variables are correlated, SHAP values may be distorted, as the method marginalises features by sampling from their marginal distribution, which assumes independence.

2.3 LIME — Local Interpretable Model-Agnostic Explanations

LIME generates local explanations by approximating a complex model with a simple linear surrogate model in the neighbourhood of a specific instance. Unlike SHAP, LIME is limited to local explanations, which may nevertheless be sufficient for many teaching use cases — understanding one student's situation rather than global model behaviour.

A Hungarian study on dropout prediction in STEM higher education[19] showed that LIME and SHAP produce complementary explanations: SHAP reveals general trends, while LIME generates more accessible, instance-specific feedback understandable by non-technical staff. The same study noted, however, that LIME's treatment of features as independent — fitting a local linear model — can lead to inaccurate explanations when non-linear dependencies exist.

2.4 Emerging Techniques

Several complementary techniques are gaining traction in educational XAI research:

- **Anchors**: generate high-precision decision rules such as "IF score below 12 AND attendance below 70% THEN high dropout risk". These IF-THEN rules are highly readable for teachers without technical training.
- **Counterfactual explanations**: indicate the minimal change needed to alter a prediction — e.g., "if this student submitted two additional assignments, their risk classification would change from at-risk to on-track". A 2024 study found counterfactuals particularly valuable for defining concrete, actionable intervention goals.[12]
- **ALE (Accumulated Local Effects)**: analyses the effect of each variable in isolation, avoiding distortions caused by correlations between features — addressing one of SHAP's key weaknesses.
- **SHAPASH**: a simplified SHAP implementation with accessible visualisation interfaces, designed for non-specialist users including educators and administrators.[20]

3. Application Domains of XAI in Education

3.1 Performance Prediction and At-Risk Student Identification

3.1.1 State of the Art

Performance prediction is the most extensively documented application of XAI in education. *Turkmen (2025)*[1], in a systematic review of 35 studies spanning 2019 to 2024, identifies it as the predominant use case of educational XAI. Models exploit diverse data sources: grades, attendance records, assignment submission timing, forum participation, and platform login patterns.

A key study by *Ujkani et al. (2024)*[13] develops a comprehensive XAI framework using Random Forest and XGBoost coupled with SHAP to predict student success or failure in online learning environments. Results consistently show that **early engagement rate** — measured during the first weeks of a course — is the single strongest predictor of final outcomes, followed by regularity of platform access. The authors emphasise that providing these explanations to educators transformed the system from a prediction tool into an actionable decision-support instrument.

A 2024 study on student adaptability[12] employing Random Forest with 91% accuracy used SHAP, LIME, Anchors, ALE, and Counterfactual explanations to reveal that "class duration" carried the highest mean SHAP value (0.175), while LIME illuminated the intricate influence of socio-economic and institutional factors. This multi-method approach demonstrated that different XAI techniques reveal different aspects of the same underlying model.

3.1.2 The Added Value of XAI over Bare Prediction

Swamy et al. (2023)[22] make a critical point: model accuracy is insufficient if teachers cannot understand the model's justifications. Their study showed that teachers significantly more often validated and acted upon AI recommendations when those recommendations were accompanied by **domain-specific explanations** rather than generic risk scores. A chemistry teacher is more persuaded by "this student struggles with stoichiometry — three of the last four errors relate to mole calculations" than by a bare probability of failure.

This finding is confirmed by *Nazaretsky et al. (2025)*[3] in an empirical study involving 41 in-service chemistry teachers who used an AI-powered recommendation tool. The study found a statistically significant positive correlation between the **understandability** of explanations, the **trust** placed in the system, and the **acceptance** of its recommendations. Two additional factors moderating adoption were identified: teachers' pedagogical perspectives and the perceived potential for workload reduction.

A stability analysis by *Tiukhova et al. (2024)*[14] added another dimension: they used SHAP not only to explain individual predictions but to investigate whether model explanations remain consistent across different student cohorts and over time. They found that stable SHAP feature importance rankings — features that matter consistently across cohorts — could be directly translated into reliable study advice for students.

3.2 Dropout Detection and Prevention

Student dropout is described in the literature as one of the most pressing issues in higher education, generating significant social and economic costs.[19] XAI-based approaches to dropout prediction offer a significant advantage over earlier statistical methods: they not only identify who is at risk, but explain why, enabling targeted rather than generic interventions.

Goran et al. (2023)[19], working with data from a large Hungarian university, demonstrated that a CatBoost classifier augmented with SHAP, permutation importance (PI), partial dependence plots (PDP), and LIME can identify STEM students at risk of dropout using only pre-enrolment achievement measures — before the student has attended a single class. The study also conducted a user survey with students and decision-makers assessing the interpretability, utility, and design quality of the visualisations, finding substantial heterogeneity in preferences.

The ICAIE (2024) review[2] synthesises XAI applications for student identification and intervention in MOOCs. It highlights that XAI enabled the detection of **early warning signals** — sudden decreases in login time, growing gaps between submissions — that teachers could not realistically monitor manually across cohorts of hundreds or thousands of students. Crucially, the review emphasises that XAI models indicating at-risk students must account for individual contextual factors such as socio-economic status, home environment, and mental health — dimensions the algorithm cannot observe.

An important equity dimension is raised by *Deho et al. (2023)*[1]: should learning analytics models include sensitive attributes (gender, ethnicity, socio-economic status) in their feature sets? Their study argues that XAI is essential precisely to examine whether such attributes are influencing predictions unduly, making it possible to detect discriminatory patterns in model behaviour.

3.3 Intelligent Tutoring Systems (ITS)

3.3.1 Personalisation and Transparency in ITS

Intelligent Tutoring Systems represent a flagship application of AI in education, providing individualised instruction and feedback. Integrating XAI into ITS addresses a fundamental paradox: the more effective a tutoring system, the more complex its internal model, and the less interpretable its recommendations — both for the learner and for the teacher.[24]

A comprehensive review of AI-based ITS[21] identifies XAI as one of the key mechanisms through which ITS can build user trust. Best practices identified include: offering accessible explanations of how decisions are made; ensuring that these explanations are interpretable by educators, students, and parents; and personalising the form of explanation to the user's background.

Conati et al. (2021)[28], in a foundational study, proposed personalising XAI explanations within ITS according to learner characteristics — "Need for Cognition", "Conscientiousness", and "Reading Proficiency". Their results showed that tailoring explanations to individual cognitive profiles significantly enhanced students' trust, perceived usefulness of the system, and learning outcomes. This concept, which they term **personalised XAI**, opens a research direction of considerable practical importance.

3.3.2 Case Study: the iRead Project

The European **iRead project**, described by *Karpouzis (2024)*[22], provides a concrete illustration of XAI in a reading acquisition ITS. The system adapts exercises in real time to each child's phonological profile and, through XAI, explains to the teacher why a specific exercise was recommended — for instance: "this child shows difficulty with consonant clusters; this exercise specifically targets that phonological pattern." Teachers participating in the project reported a marked reduction in anxiety about AI-assisted teaching and better integration of the tool into their practice. The explainability component was credited with alleviating the "black box" concern that had deterred adoption in earlier trials.

3.4 Automated Essay Scoring

Automated Essay Scoring (AES) is a high-stakes application of educational AI where explainability is not merely desirable but ethically necessary. A grade assigned without explanation provides little formative value to the learner, and a purely algorithmic score raises legitimate concerns about accountability.

Boulanger and Kumar (2020)[1] developed a system called **SHAPed Essay Scoring** that uses SHAP to decompose a writing score into individual contributions from linguistic features: discourse coherence, vocabulary richness, argument structure, syntactic complexity. This allows both the teacher and the student to understand which dimensions of writing are contributing positively or negatively to the overall assessment.

Conijn et al. (2023)[1], publishing in the *Journal of Learning Analytics*, demonstrated in a controlled study that explanations provided by AES systems significantly increased student **trust** in the system and **motivation** to revise their work — but only when the explanations were perceived as both comprehensible and fair. Explanations that were technically correct but opaque to students had no motivational effect.

3.5 Adaptive Learning and Resource Recommendation

Recommendation systems that suggest learning resources, exercises, or next steps based on a student's profile and history are increasingly common in educational platforms. Without XAI, these recommendations are perceived as arbitrary. *Huang et al. (2023)*[22] showed that recommendations accompanied by explanations (e.g., "we recommend this exercise because you scored 65% on synthesis questions in the last three assignments") increased student engagement and improved outcomes in a flipped classroom setting.

A significant limitation is flagged by *Bulathwela et al. (2024)*[1]: AI-driven personalised recommendation systems tend to disproportionately benefit moderately motivated learners, leaving both the least and most motivated students underserved. XAI can help identify this systemic bias by making the recommendation logic transparent, enabling targeted recalibration. Their broader warning — "artificial intelligence alone will not democratise education" — serves as an important counterbalance to techno-optimist narratives.

4. Ethical Challenges, Fairness, and Limitations

4.1 Algorithmic Bias in Education

Predictive models trained on historical data risk reproducing and amplifying pre-existing inequalities. If training data reflects systemic biases — for instance, students from disadvantaged socio-economic backgrounds historically achieving lower grades due to structural barriers — the model will learn to predict failure for these groups, creating a **self-fulfilling prophecy**.[6]

XAI plays a crucial role here: by rendering visible the variables that influence predictions, it enables the detection of whether sensitive attributes (gender, ethnicity, socio-economic status) are inappropriately influencing algorithmic judgements. *Deho et al. (2023)*[1] argue that XAI must at minimum allow explanation of the impact of sensitive attributes, so that decisions remain ethically defensible. They stop short of recommending automatic exclusion of sensitive attributes, noting that this can paradoxically reinforce bias by preventing its detection.

4.2 The FATE Framework

Educational AI research has adopted the **FATE** framework — Fairness, Accountability, Transparency, Ethics — as an evaluation grid for AI systems in education.[26] Within this framework, XAI primarily contributes to Transparency and underpins Accountability, enabling institutions to justify automated decisions affecting students.

The European Commission's ethical guidelines on AI in education[29] explicitly call for transparency and accountability in AI-supported educational interventions. The EU AI Act (2024) classifies educational AI systems as **high-risk**, imposing transparency requirements that render XAI not merely desirable but legally mandated for certain deployment contexts. Institutions deploying AI-assisted assessment, student profiling, or admission tools within the EU are subject to these requirements.

4.3 Key Limitations and Caveats

The research literature identifies several important limitations that teachers and institutions must be aware of:

- **The accuracy-explainability trade-off:** The most accurate models (deep neural networks) are generally the least transparent. Choosing an intrinsically interpretable model may mean accepting lower predictive performance.[7]
- **Confirmation bias:** XAI explanations can reinforce a teacher's pre-existing judgements by seemingly validating them with data, even when those judgements are incorrect. Bertrand et al. (2022) showed that cognitive biases significantly affect XAI-assisted decision-making.[22]
- **Insufficient human evaluation:** A 2025 MIT study found that only 0.7% of XAI papers across 18,000 publications performed any human evaluation to verify whether explanations were actually understood by their intended audience.[9] This is a remarkable gap between the claimed explainability of systems and evidence of their comprehensibility.

- **Over-reliance risk:** Teachers may delegate their professional judgement to an algorithm precisely because it "explains itself", without exercising critical scrutiny. Explainability does not guarantee correctness.
- **Absent contextual data:** Algorithmic models cannot observe extra-curricular factors — family difficulties, mental health, recent trauma — that a teacher in direct contact with a student can perceive. XAI outputs should always be contextualised by qualitative human knowledge.[2]
- **Population mismatch:** A model trained on data from one student population may produce unreliable and potentially biased explanations when applied to a different population. Stability analysis across cohorts, as demonstrated by Tiukhova et al. (2024), is essential before broad deployment.[14]

5. Recommendations for Teachers

The following ten recommendations are grounded in the empirical findings synthesised above. They are organised around three axes: **understanding and evaluating** XAI tools; **using them in an informed manner**; and **preserving professional judgement**. They are addressed both to individual teachers and to institutions.

Recommendation 1: Develop AI and XAI Literacy

- Acquire a working understanding of how machine learning algorithms function — not to become a developer, but to know which questions to ask of an AI-powered tool.
- Understand the distinction between correlation and causation: a model that predicts failure from low attendance does not prove that attendance causes failure.
- Participate in institutional training sessions on AI-powered educational tools.
- Consult accessible resources: UNESCO guidelines on ethical AI in education, introductory MOOCs on algorithms and data literacy.

Recommendation 2: Demand Transparency from EdTech Tools

- Before adopting an AI educational tool, systematically ask: "Does this tool explain its decisions and recommendations, and if so, how?"
- Verify whether the tool provides local explanations (why this recommendation for this student?) and not only aggregate statistics.
- Prioritise tools that explicitly disclose the explainability techniques used (SHAP, LIME, decision trees, rule-based systems).
- Flag to your institution any tool making high-impact decisions (tracking, grading, progression) without providing readable justifications.

Recommendation 3: Treat Predictions as Hypotheses, Not Verdicts

- Consider AI-generated risk alerts as hypotheses to be validated, not definitive diagnoses about a student's trajectory.
- Systematically cross-reference algorithmic predictions with direct classroom observation and direct dialogue with the student.
- Document cases where your professional observation diverges from the prediction — these cases are valuable for identifying model biases.
- In online or MOOC contexts, supplement automatic engagement indicators with qualitative check-ins that the algorithm cannot replicate.

Recommendation 4: Tailor Interventions to the Explanations Provided

- When SHAP identifies "regularity of logins" as the primary risk factor for a specific student, design an intervention targeting that factor directly (weekly check-ins, structured reminders, paired study groups).

- Use counterfactual explanations ("if this student submitted two additional assignments, their risk classification would change") to set concrete, measurable goals with students.
- Share explanations with students where appropriate — research shows that transparency about risk factors increases student self-regulation and motivation.[1]
- Avoid generic "at-risk" interventions triggered by a bare probability score without reading the underlying explanatory factors.

Recommendation 5: Actively Monitor for Algorithmic Bias

- Check whether risk alerts are distributed homogeneously across student subgroups (gender, socio-economic status, ethnicity, first-generation students).
- If a particular group is systematically over-represented in negative predictions, raise the issue with your institution or the tool's vendor.
- Advocate for regular bias audits of deployed AI systems, with results made available to teaching staff.
- Be especially cautious with tools trained on data from populations substantially different from your own student body.

Recommendation 6: Involve Students in Understanding the Tools

- Explain to students, in broad terms, which AI tools are used in their course and how they function.
- Share relevant XAI outputs with students where appropriate: "according to the analytics system, your engagement in discussion forums is a strong positive factor for your likely success."
- Research demonstrates that transparency with students increases trust and motivation, provided explanations are perceived as fair and comprehensible.[1]
- Use AI-related discussions as opportunities for critical thinking about data, algorithms, and digital ethics.

Recommendation 7: Maintain the Primacy of Professional Judgement

- Explainable AI remains a decision-support tool — the final decision belongs to the teacher.
- Resist the over-reliance trap: a comprehensible explanation is not necessarily a correct one. Apply the same critical scrutiny to AI explanations that you would to any source of information.
- Preserve and value the contextual knowledge that algorithms cannot access: family dynamics, mental health, recent events in a student's life.
- Exercise your right to override an algorithmic recommendation and document your reasoning — this is an essential form of professional regulation and accountability.

Recommendation 8: Participate in the Co-Design of XAI Tools

- Respond to user surveys from EdTech vendors by specifically emphasising the comprehensibility (or lack thereof) of the explanations provided.

- Participate in pilot programmes and research partnerships that involve teachers in the design of XAI interfaces.
- Articulate your real decision-making needs: which pedagogical decisions would you like AI to help with? What form of explanation would be genuinely actionable in your practice?
- Research by Nazaretsky et al. (2025) demonstrates that domain-specific explanations — tailored to the subject matter being taught — are significantly more effective than generic statistical outputs.[3]

Recommendation 9: Advocate for Institutional Governance and Policy

- Advocate at the institutional level for the adoption of an algorithmic transparency policy, in line with the EU AI Act (2024) requirements.
- Seek contract provisions with EdTech vendors specifying the explainability techniques deployed and guarantees of non-discrimination.
- Support the establishment of ethics committees including teachers, students, parents, and AI experts to oversee deployments.
- Limit high-stakes decisions (tracking, grade appeals, progression) to contexts where verifiable XAI explanations are available and have been reviewed.

Recommendation 10: Stay Informed as the Field Rapidly Evolves

- Follow publications from specialised workshops: HEXED (Human-Centric eXplainable AI in Education), LAK (Learning Analytics and Knowledge), AIED (AI in Education).
- Monitor policy developments: EU AI Act implementation, UNESCO recommendations on AI and education, national digital education strategies.
- Collaborate with university research groups working on educational XAI to access findings before they reach mainstream EdTech products.
- Share experiences and practices within professional teacher networks — collective knowledge about which tools explain themselves well (or poorly) is a public good.

6. Future Research Directions

6.1 Generative AI and LLMs

The emergence of large language models (LLMs) in educational settings raises new explainability challenges. *Susnjak (2024)*[10] proposes approaches combining XAI with LLMs to generate not only predictions but contextualised, natural language intervention plans — what he terms "prescriptive analytics with ChatGPT". The explainability of LLMs themselves (why did the model generate this feedback?) remains a largely open research problem, and the emerging field of generative XAI (GenXAI) is only beginning to address it.

6.2 Multimodal XAI

Recent research is exploring XAI applied to multimodal data: facial expressions, posture, physiological signals, and eye-tracking patterns.[10] These approaches could theoretically detect engagement or confusion states invisible in platform log data. They raise, however, profound ethical questions about student surveillance that must be addressed before any deployment, and the explainability of multimodal models remains technically challenging.

6.3 Standardisation and Human Evaluation

The MIT finding that only 0.7% of XAI papers include human evaluations[9] constitutes a call for urgent methodological reform. Metrics such as **fidelity** (does the explanation reflect the model's true reasoning?), **completeness**, and **comprehensibility** (as measured by human participants) should become mandatory criteria in academic publications and EdTech certification frameworks. Without such standards, "explainable AI" risks becoming a marketing label rather than a substantive guarantee.

6.4 Equity-Centred XAI Frameworks

Bulathwela et al. (2024)[1] remind us that "AI alone will not democratise education." XAI must be embedded in a broader equity framework that addresses structural inequalities algorithms may reproduce. The emerging PEARL framework (Designing Understandable and Fair AI for Learning, 2026)[30] attempts to integrate fairness and comprehensibility guarantees from the design stage, representing a promising direction for human-centred educational AI.

7. Conclusion

Explainable Artificial Intelligence represents far more than a technical refinement of educational AI systems — it constitutes a **pedagogical and ethical necessity**. As Saqr and Lopez-Pernas (2025)[10] state, enabling users to scrutinise and refine AI decisions is a necessary step in aligning technological advancement with the ethical and practical needs of education.

This documentation has examined five major application domains — performance prediction, dropout prevention, intelligent tutoring systems, automated essay scoring, and adaptive learning — and has documented, across each, the added value of XAI over opaque AI: improved teacher trust, more targeted pedagogical interventions, greater student motivation, and enhanced capacity to detect and correct algorithmic bias.

The ten recommendations formulated aim to equip teachers as critical, informed users of XAI-powered tools — able to leverage their power while preserving professional judgement and protecting students from the risks of algorithmic opacity and bias. The central challenge ahead is not technical but institutional and cultural: building educational ecosystems in which AI amplifies — rather than replaces — the irreducible intelligence of human teaching.

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